Khazar Ahmadi

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Research Overview

My research centers on uncovering the neural underpinnings of perception, memory and high-level cognitive functions across the spectrum of health, aging, and disease. To achieve this, I employ a multidisciplinary approach that integrates state-of-the-art neuroimaging, behavioral paradigms, and computational modeling to unravel the fine-grained functional and structural architecture of the human brain.

Academic Experience

Ruhr University Bochum — *Bochum, Germany*
Postdoctoral Researcher with Prof. Nikolai Axmacher — 2022 - 2026

National Institutes of Health (NIH) — *Bethesda, USA*
Visiting Researcher with Dr. Renzo Huber — 2024

Lund University — *Lund, Sweden*
Postdoctoral Researcher with Prof. Oskar Hansson — 2020 - 2022

Spinoza Center for Neuroimaging — *Amsterdam, Netherlands*
Visiting Researcher with Prof. Serge Dumoulin — 2017

University Medical Center Groningen — *Groningen, Netherlands*
Visiting Researcher with Prof. Frans Cornelissen — 2017

Education

Otto-von-Guericke University — *Magdeburg, Germany*
PhD in psychology (magna cum laude) — 2014 - 2019

- Thesis title: Plasticity and stability of the cortical wiring in the human visual system

Shahid Beheshti University — *Tehran, Iran*
M.Sc. in child and adolescent clinical psychology (with distinction) — 2010 - 2013

Tabriz University — *Tabriz, Iran*
B.Sc. in clinical psychology (with distinction) — 2006 - 2010

Publications

Submitted Manuscripts

- Akbari, A., **Ahmadi, K.**, Saligheh-rad, H.R., Guidi, M. (2026). VASO fMRI: From Contrast Mechanism to Application in Neuroscience. [Review article]
- Rau, E.M.B., Heinen, R., Herweg, N.A., Patai, E.Z., Kobelt, M., **Ahmadi, K.**, Axmacher, N. (2025). Task-relevant cognitive maps in episodic memory. bioRxiv. [pre-print link]

Published Manuscripts

- **Ahmadi, K.**, Swegle, S., Kashyap, S., Bouyeure, A., Bandettini, P., Axmacher, N., Huber, R. (2026). Blood volume sensitive laminar fMRI with VASO in human hippocampus: Capabilities and biophysical challenges at clinical 7T scanners. Imaging Neuroscience. [Link]
- Anijärv, T., Ossenkoppele, R., Smith, R., Binette, A.P., Collij, L., Behjat, H., Rittmo, J., Karlsson, L., **Ahmadi, K.**, Strandberg, O., van Westen, D., Vogel, J., Stomrud, E., Palmqvist, S., Mattsson-Carlgren, N., Spotorno, N., Hansson, O. (2025). Hemispheric asymmetry of tau pathology is related to asymmetric amyloid deposition in Alzheimer's disease. Nature Communications. [Link]

- **Ahmadi, K.**, Pereira, J., van Westen, D., Pasternak, O., Zhang, F., Nilsson, Stomrud, E., Spotorno, N. and Hansson, O. (2024). Fixel-based analysis reveals tau-related white matter changes in early stages of Alzheimer's disease. Journal of Neuroscience. [Link]
- Molz, B., Herbik, A., Baseler, H., de Best, P., RAZ, N., Gouws, A., **Ahmadi, K.**, Lowndes, R., McLean, R., Gottlob, I., Kohl, S., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Morland, A., and Hoffmann, M.B. (2023). Achromatopsia - limits to visual cortex plasticity in the absence of functional cones. Investigative Ophthalmology & Visual Science [Link]
- Cicognola, C., Mattsson-Carlgren, N., van Westen, D., Zetterberg, H., Blennow, K., Palmqvist, S., **Ahmadi, K.**, Strandberg, O., Stomrud, E., Janelidze, S., Hansson, O. (2023). Associations of CSF PDGFRβ with aging, blood-brain barrier damage, neuroinflammation and Alzheimer disease pathological changes. Neurology. [Link]
- **Ahmadi, K.**, Pereira, J. B., Berron, D., Vogel, J., Ingala, S., Strandberg, O. T., Janelidze, S., Barkhof, F., Pfeuffer, J., Knutsson, L., van Westen, D., Palmqvist, S., Mutsaerts, H. J., Hansson, O. (2022). Gray matter hypoperfusion is a late pathological event in the course of Alzheimer's disease. Journal of Cerebral Blood Flow & Metabolism. [Link]
- Molz, B., Herbik, A., Baseler, H. A., de Best, P., Vernon, R., Raz, N., Gouws, A. D., **Ahmadi, K.**, Lowndes, R., Mclean, R. J., Gottlob, I., Kohl, S., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Hoffmann, M. B., Morland, A. B. (2021). Structural changes to primary visual cortex in the congenital absence of cone input in achromatopsia. NeuroImage: Clinical. [Link]
- Puzniak, R. J., McPherson, B.*, **Ahmadi, K.***, Herbik, A., Kaufmann, J., Liebe, T., Gouws, A. D., Morland, A. B., Gottlob, I., Hoffmann, M. B., Pestilli, F. (2021). CHIASM, the human brain albinism and achiasma MRI dataset. Scientific Data. [* shared second-authorship; Link]
- Lowndes, R., Molz, B., Warriner, L., Herbik, A., de Best, P., Raz, N., Gouws, A. D., **Ahmadi, K.**, Mclean, R. J., Gottlob, I., Kohl, S., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Hoffmann, M. B., Morland, A. B., Baseler, H. (2021). Structural differences across multiple visual cortical regions in the absence of cone function in congenital achromatopsia. Frontiers in Neuroscience. [Link]
- **Ahmadi, K.**, Fracasso, A., Puzniak, R. J., Gouws, A. D., Yakupov, R., Speck, O., Kaufmann, J., Pestilli, F., Dumoulin, S. O., Morland, A. B., Hoffmann, M. B. (2020).Triple visual hemifield maps in optic chiasm hypoplasia. NeuroImage. [Link]
- Carvalho, J., Invernizzi, A., **Ahmadi, K.**, Hoffmann, M. B., Renken, R. J., Cornelissen, F. W. (2020).Micro-probing enables fine-grained mapping of neuronal populations using fMRI. NeuroImage. [Link]
- Puzniak, R. J., **Ahmadi, K.**, Kaufmann, J., Gouws, A. D., Morland, A. B., Pestilli, F., Hoffmann, M. B. (2019). Quantifying nerve decussation abnormalities in the optic chiasm. NeuroImage: Clinical. [Link]
- **Ahmadi, K.**, Herbik, A., Wagner, M., Kanowski, M., Thieme, H., Hoffmann, M. B. (2019). Population receptive field and connectivity properties of the early visual cortex in human albinism. NeuroImage. [Link]
- Eick, C., **Ahmadi, K.**, Sweeney-Reed, C. M., Hoffmann, M. B. (2019). Interocular transfer of visual memory: influence of visual impairment and abnormalities of the optic chiasm. Neuropsychologia. [Link]
- **Ahmadi, K.**, Fracasso, A., van Dijk, J. A., Kruijt, C., van Genderen, M., Dumoulin, S. O., & Hoffmann, M. B. (2018). Altered organization of the visual cortex in FHONDA syndrome. NeuroImage. [Link]
- Arngrim[+], N., Hougaard[+], A., **Ahmadi, K.***, Vestergaard*, M. B., Schytz, H. W., Amin, F. M., Larsson, H.B.W., Olesen, J., Hoffmann, M.B. & Ashina, M. (2017). Heterogenous migraine aura symptoms correlate with visual cortex functional magnetic resonance imaging responses. Annals of Neurology. [Link]
- Hoffmann, M. B., Thieme, H., & **Ahmadi, K.** (2017). Potential of fMRI for the Functional Assessment of the Pathological Visual System. Klinische Monatsblätter für Augenheilkunde. [Link]
- **Ahmadi, K.**, Pouretemad, H., Esfandiari, J., Yoonessi, A. (2015). Psychophysical evidence for impaired magno, parvo, and konio-cellular pathways in dyslexic children. Journal of Ophthalmic & Vision Research. [Link]

Conference Presentations

Talks

- **Ahmadi, K.** (2026). Enhancing fMRI specificity to neural activity using VASO and capillary blood volume imaging. European Society for Magnetic Resonance in Medicine and Biology (ESMRMB), Girona, Spain.

- **Ahmadi, K.** (2025). BOLD and non-BOLD contrasts for laminar fMRI at ultra-high field in human hippocampus. Spring Hippocampal Research Conference, Verona, Italy.
- **Ahmadi, K.** (2024). Laminar profile of hippocampal subregions during spatial navigation. International Society for Magnetic Resonance in Medicine (ISMRM), Singapore, Singapore.
- **Ahmadi, K.** (2022). Tau accumulation is associated with fiber-specific white matter degeneration in Alzheimer's disease. International Society for Magnetic Resonance in Medicine (ISMRM), London, England.
- **Ahmadi, K.** (2022). Disentangling fiber-specific white matter alterations in relation to hallmarks of Alzheimer's disease. Alzheimer's Disease and Parkinson's Disease (ADPD), Barcelona, Spain.
- **Ahmadi, K.** (2021). Altered cerebral blood perfusion in Alzheimer's disease spectrum and its association with amyloid-β and tau pathology. Alzheimer's Disease and Parkinson's Disease (ADPD), Virtual conference.
- **Ahmadi, K.** (2018). Macroscopic and mesoscopic cortical organization in congenital visual pathway abnormalities. Brain-in-depth (BID) - Symposium on layer-dependent MRI, German Center for Neurodegenerative Diseases (DZNE), Magdeburg, Germany.
- **Ahmadi, K.** (2016). Altered retino-cortical connections and visual cortex reorganization in the recently discovered FHONDA syndrome. Society for Neuroscience (SfN), San Diego, USA.

Poster Presentations

- Schneider, J., Genc, E., **Ahmadi, K.**, Axmacher, N., Kumsta, R. (2026). Multimodal signatures of cross-disorder polygenic risk and structural connectivity are linked to negative affect. Cognitive Science Society (CSS). Rio de Janeiro, Brazil. [Abstract Link]
- **Ahmadi, K.**, Swegle, S., Rubin, M., Morgan, A. T., Bouyeure, A., Bandettini, P., Axmacher, N., Huber, R. (2025). Blood volume sensitive laminar fMRI with VASO in human hippocampus: Capabilities of clinical 7T and biophysical challenges. International Society for Magnetic Resonance in Medicine (ISMRM), Hawaii, USA. [YouTube Link]
- **Ahmadi, K.**, Stawarczyk, D., Pfaffenrot, V., Gomes, C. A., Kashyap, S., Patai, Z., Norris, D. G., Axmacher, N. (2024). Laminar profiles of hippocampal subfields are differentially associated with navigation strategies. Organization for Human Brain Mapping (OHBM), Seoul, south Korea.
- Rau, E. M. B., Herweg, N. A., Heinen, R., **Ahmadi, K.**, Axmacher, N. (2024). Functions of the medial temporal lobe in memory and navigation of abstract conceptual spaces. Federation of European Neuroscience Societies (FENS), Vienna, Austria.
- Kashyap, S., Gomes, C. A., **Ahmadi, K.**, Axmacher, N., Uludag, K. (2024). A new proposed workflow for processing high spatial resolution MRI data with multiple sessions. Cognitive Computational Neuroscience (CCN), Boston, USA.
- **Ahmadi, K.**, Pereira, J. B., Berron, D., Vogel, J., Ingala, S., Strandberg, O. T., Janelidze, S., Barkhof, F., Pfeuffer, J., Knutsson, L., van Westen, D., Mutsaerts, H. J., Palmqvist, S., Hansson, O. (2021). Tau and synaptic biomarkers but not amyloid-β are associated with cerebral perfusion in the Alzheimer's disease spectrum. Alzheimer's Association International Conference (AAIC), Virtual conference.
- Cicognola, C., Janelidze, S., Van Westen, D., **Ahmadi, K.**, Hansson, O. (2021). Cerebrospinal fluid platelet-derived growth factor receptor beta measured in BioFinder-2: an early biomarker in the Alzheimer's disease continuum? Clinical Trials on Alzheimer's Disease (CTAD), Boston, USA.
- Puzniak, R. J., McPherson, B., **Ahmadi, K.**, Herbik, A., Kaufmann, J., Liebe, T., Gouws, A. D., Morland, A. B., Gottlob, I., Hoffmann, M. B., Pestilli, F. (2020). Chiasmal malformations dataset: a unique neuroimaging testbed. Vision Science Society (VSS), Virtual conference.
- Hoffmann, M. B., Molz, B., Herbik, A., de Best, P., Raz, N., Gouws, A. D., **Ahmadi, K.**, Lowndes, R., McLean, R. J., Kohl, S., Gottlob, I., Choritz, L., Maguire, J., Kanowski, M., Käsmann-Kellner, B., Wieland, I., Banin, E., Levin, N., Baseler, H. Morland, A. B. (2020). Visual cortex stability and plasticity in the absence of functional cones in achromatopsia. Vision Science Society (VSS), Virtual conference.

- **Ahmadi, K.**, Fracasso, A., Puzniak, R. J., Gouws, A. D., Yakupov, R., Speck, O., Kaufmann, J., Pestilli, F., Dumoulin, S. O., Morland, A. B., Hoffmann, M. B. (2019). Triple hemi-field input to the visual cortex in a patient with chiasmal hypoplasia. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Carvalho, J., **Ahmadi, K.**, Invernizzi, A., Hoffmann, M. B., Renken, R., Cornelissen, F. W. (2019). Micro-probing the visual cortex: high-resolution mapping of neuronal sub-populations. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Puzniak, R. J., **Ahmadi, K.**, Kaufmann, J., Gouws, A. D., Morland, A. B., Pestilli, F., Hoffmann, M. B. (2019). Quantification of nerve decussation abnormalities in the optic chiasm. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- Prabhakaran, G., Al-Nosairy, K. O., Tempelmann, C., **Ahmadi, K.**, Hoffmann, M. B. (2019). Brain activity in input-deprived visual cortex in glaucoma – no fMRI-evidence of plasticity. Organization for Human Brain Mapping (OHBM), Rome, Italy.
- **Ahmadi, K.**, Fracasso, A., Gouws, A. D., Morland, A. B., Dumoulin, S. O., Hoffmann, M. B. (2017). Functional organization of the visual cortex in a unique case of achiasma. Society for Neuroscience (SfN), Washington, D.C., USA.
- **Ahmadi, K.**, Fracasso, A., van Dijk, J. A., Kruijt, C., van Genderen, M., Dumoulin, S. O., & Hoffmann, M. B. (2017). Cortical plasticity in FHONDA: a new inherited visual system disorder. European conference on visual perception (ECVP), Berlin, Germany.
- **Ahmadi, K.**, Pouretemad, H. R., Esfandiari, J., Yoonessi, A. (2013). Investigation of magno, parvo and koniocellular pathways in children with dyslexia. International Conference of Cognitive Sciences (ICCS), Tehran, Iran.

Technical Skills

Neuroimaging Modalities	fMRI (at 7 and 3 T), dMRI, sMRI & ASL
Neuroimaging Software	Nipype, AFNI, FSL, MRtrix, DIPY, FreeSurfer, LayNii, ExploreASL, HippUnfold, ASHS, SPM, ANTs & MrVista
Neuroimaging Hardware	Siemens 7T MAGNETOM Terra Scanner Operator
Programming & Version Control	Bash, R, Python, MATLAB, GitHub, LaTeX
Experiment Software	PsychoPy, Psychtoolbox, Neurobehavioral Systems Presentation
Languages	Azerbaijani (native), Persian (native), English (fluent), German (upper-intermediate fluency)

Awards and Honors

Marie-Sklodowska Curie Actions (MSCA) Seal of Excellence *2026*
Postdoctoral fellowship score 94%, funding threshold at 97% — European Union

Finalist, ISMRM trainee poster award *2025*
Selected among the top five abstracts for the poster award competition at the ISMRM High Field Study Group — Hawaii, USA

Winner, Erwin L. Hahn Institute's brain-art competition *2023*
Awarded € 200 for the art-work "Hippocampal Moon" — Essen, Germany

Greta & Johan Kock Foundation grant for medical research *2021*
(About) — Lund, Sweden
Awarded € 10,000 for the project "Clinical Use of Novel dMRI Markers in Alzheimer's Disease"

Lund University MultiPark travel grant *2021-2022*
(About) — Lund, Sweden
Awarded a total of € 2,000, covering conference registrations and travel expenses

The ISMRM Summa Cum Laude Merit Award *2022*
(About) London, UK

- Selected abstract in top 5% of all submissions

Marie Curie Ph.D Fellowship *2015-2018*
Research Training in computational Visual Neuroscience Magdeburg, Germany

Top Student Award *2010*
Ranked 1st out of 50 students based on cumulative GPA in BSc program, received a cash prize of € 100 Tabriz, Iran

Teaching & Mentorship

Ruhr University Bochum *Bochum, Germany*
Lead instructor, "fMRI: from Theory to Practice" 2022 - 2026
Lead instructor, "Aging, dementia and memory disorders" 2024
Mentored students:

- Dana Schröder & Paul Höchter (Research assistants)
- Kübra Karatas & Niko Britt (MSc. & BSc. Interns)
- Alena Machentanz (BSc. thesis title: "Memory performance in active vs. passive learning contexts")

Neuromatch Academy: Computational Neuroscience Workshop *Online Event*
Mentor of the group "NeuroMatched Philosophers" 2021

- Project: predicting fearful stimuli from HCP fMRI data using MVPA

Otto-von-Guericke University *Magdeburg, Germany*
Instructor: "Neuroscience Journal Club" 2016 - 2018
Co-mentored students:

- Charlotta Eick (MSc. thesis title: "Inter-ocular transfer of visual memory in albinism")
- Gokulraj Prabhakaran & Anne Herbik (PhD student & postdoc)

Professional Service

Ad-hoc Reviewer
NeuroImage, Cell Reports Medicine, Alzheimer's & Dementia, Alzheimer's Research & Therapy, Journal of Cerebral Blood Flow & Metabolism, Brain Communications, Scientific Reports, Neuropsychologia, Brain Research Bulletin, Journal of Neuroscience Methods, British Journal of Ophthalmology

Symposia & Lecture Organization *Bochum, Germany*

- Advances in Layer-specific fMRI
- Surface-based analysis for hippocampal subfield segmentation 2023

Public Outreach *Magdeburg, Germany*

- Long Night of Science at Otto-von-Guericke University 2016 - 2019

Oslo, Norway

- Popular-science talk at 4th European Days of Albinism 2018